

Effects of Polyphosphate Additives on Campylobacter Survival in Processed Chicken Exudates

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Campylobacter spp. are responsible for a large number of the bacterial food poisoning cases worldwide. Despite being sensitive to oxygen and nutritionally fastidious, Campylobacter spp. are able to survive in food processing environments and reach consumers in sufficient numbers to cause disease. To investigate Campylobacter persistence on processed chicken, exudates from chickens produced for consumer sale were collected and sterilized. Two types of exudates from chicken products were collected: enhanced, where a marinade was added to the chickens during processing, and nonenhanced, where no additives were added during processing. Exudates from enhanced chicken products examined in this study contained a mixture of polyphosphates. Exudate samples were inoculated with Campylobacter jejuni or Campylobacter coli strains and incubated under a range of environmental conditions, and viable bacteria present in the resultant cultures were enumerated. When incubated at 42°C in a microaerobic environment, exudates from enhanced chicken products resulted in increased survival of C. jejuni and C. coli compared with that in nonenhanced exudates in the range of <1 to >4 log CFU/ml. Under more relevant food storage conditions (4°C and normal atmosphere), the exudates from enhanced chicken products also demonstrated improved Campylobacter survival compared with that in nonenhanced exudates. Polyphosphates present in the enhanced exudates were determined to be largely responsible for the improved survival observed when the two types of exudates were compared. Therefore, polyphosphates used to enhance chicken quality aid in sustaining the numbers of Campylobacter bacteria, increasing the opportunity for disease via cross-contamination or improperly cooked poultry.